Instructions: Show your work. Use a ruler and protractor when a drawing or measurement is requested. Unless a diagram is provided, sketch one clearly.

Practice 1: Points, lines, and segments

Name: ____________________ Date: __________

1. Explain the difference between a line, a line segment, and a ray. Give one example of each.

Work space

2. Draw segment $\overline{AB}$, ray $\overrightarrow{CD}$, and line $\overleftrightarrow{EF}$. Mark each endpoint.

Work space

Practice 2: Parallel and perpendicular lines

Name: ____________________ Date: __________

1. Draw a pair of parallel lines and a pair of perpendicular lines. Label the right angle in your perpendicular pair.

Work space

2. A rectangle has sides AB and CD opposite each other. Are they parallel, perpendicular, or both? Explain.

Work space

Practice 3: Classifying angles

Name: ____________________ Date: __________

1. Classify each angle as acute, right, obtuse, or straight: $35°$, $90°$, $125°$, and $180°$.

Work space

2. Sketch one angle of each type from problem 1 and label its measure.

Work space

Practice 4: Measuring and drawing angles

Name: ____________________ Date: __________

1. Use a protractor to draw an angle measuring $65°$. Label the vertex and rays.

Work space

2. Draw an angle measuring $140°$. Is it acute, right, or obtuse? Explain how you know.

Work space

Practice 5: Angle relationships

Name: ____________________ Date: __________

1. Two angles form a straight angle. One measures 72°. Find the other angle.

Work space

2. A right angle is split into angles measuring x° and 37°. Find x.

Work space

Practice 6: Triangles by sides

Name: ____________________ Date: __________

1. Classify a triangle with side lengths 6 cm, 6 cm, and 6 cm. Explain.

Work space

2. Classify a triangle with side lengths 5 in, 5 in, and 8 in. Explain.

Work space

Practice 7: Triangles by angles

Name: ____________________ Date: __________

1. Classify a triangle whose angles are 45°, 45°, and 90°.

Work space

2. A triangle has angles 30°, 70°, and 80°. Is it acute, right, or obtuse? Show why.

Work space

Practice 8: Quadrilaterals

Name: ____________________ Date: __________

1. Name the quadrilateral with four equal sides and four right angles. List two of its properties.

Work space

2. A quadrilateral has exactly one pair of parallel sides. What is it called? Draw an example.

Work space

Practice 9: Polygons and attributes

Name: ____________________ Date: __________

1. A polygon has 7 sides. What is its name? How many vertices does it have?

Work space

2. Sort these figures by number of sides: triangle, pentagon, hexagon, octagon. Write the side count for each.

Work space

Practice 10: Perimeter of polygons

Name: ____________________ Date: __________

1. Find the perimeter of a rectangle measuring 12 cm by 7 cm. Show the formula.

Work space

2. A regular pentagon has sides of 9 m each. Find its perimeter.

Work space

Practice 11: Area of rectangles and squares

Name: ____________________ Date: __________

1. Find the area of a rectangle that is 14 ft long and 6 ft wide. Include square units.

Work space

2. A square has side length 11 in. Find its area and perimeter.

Work space

Practice 12: Area of composite rectangles

Name: ____________________ Date: __________

1. An L-shape can be split into rectangles of 8 m by 3 m and 4 m by 5 m. Find its total area.

Work space

2. A rectangle 10 units by 9 units has a corner 3 units by 4 units cut out. Find the remaining area.

Work space

Practice 13: Coordinate plane basics

Name: ____________________ Date: __________

1. Draw a coordinate grid with both axes numbered from 0 to 6 in steps of 1. Plot and label $A(2,4)$, $B(6,4)$, and $C(6,1)$. Which points share a horizontal line?

Work space

2. Write the ordered pair for a point 5 units right and 3 units up from the origin.

Work space

Practice 14: Coordinate distances and shapes

Name: ____________________ Date: __________

1. Points $P(1, 2)$ and $Q(7, 2)$ are on a coordinate grid. Find the length of $\overline{PQ}$ in units.

Work space

2. Draw a coordinate grid with both axes numbered from 0 to 6 in steps of 1. Plot $(1, 1)$, $(1, 5)$, $(6, 5)$, and $(6, 1)$. Connect them in order, then connect the last point to the first. Name the shape and find its perimeter.

Work space

Practice 15: Lines of symmetry

Name: ____________________ Date: __________

1. Draw a line of symmetry for a square. How many different lines of symmetry does a square have?

Work space

2. How many lines of symmetry does a non-square rectangle have? Explain by describing a fold.

Work space

Practice 16: Symmetry in polygons

Name: ____________________ Date: __________

1. How many lines of symmetry does an equilateral triangle have? Sketch them.

Work space

2. How many lines of symmetry does a regular pentagon have? State the rule you used.

Work space

Practice 17: Reflection on a grid

Name: ____________________ Date: __________

1. Draw a coordinate grid with both axes numbered from 0 to 6 in steps of 1. Draw a vertical fold line through $(4, 0)$ and $(4, 6)$. Point $(3, 2)$ is 1 unit left of this line. Mark its mirror image 1 unit to the right of the line, at the same height. Give the new ordered pair and explain the change.

Work space

2. Draw a coordinate grid with both axes numbered from 0 to 6 in steps of 1. Draw a horizontal fold line through $(0, 3)$ and $(6, 3)$. Point $(4, 1)$ is 2 units below this line. Mark its mirror image 2 units above the line, directly above the original point. Give the new ordered pair.

Work space

Practice 18: Classify and justify

Name: ____________________ Date: __________

1. A quadrilateral has 4 equal sides, but its angles are not right angles. Is it a square? Name the shape and justify.

Work space

2. A quadrilateral has four right angles and opposite sides equal. What shape is it? Could it also be a square? Explain.

Work space

Practice 19: Geometry problem solving

Name: ____________________ Date: __________

1. A rectangular garden is 18 m by 9 m. Find its perimeter and area.

Work space

2. A regular hexagon has perimeter 48 cm. Find the length of one side.

Work space

Practice 20: Mixed review

Name: ____________________ Date: __________

1. An angle measures 90°. Name its type, then draw a pair of perpendicular lines that make it.

Work space

2. A rectangle is 8 units by 5 units. Find its area and perimeter, and state how many lines of symmetry it has.

Work space